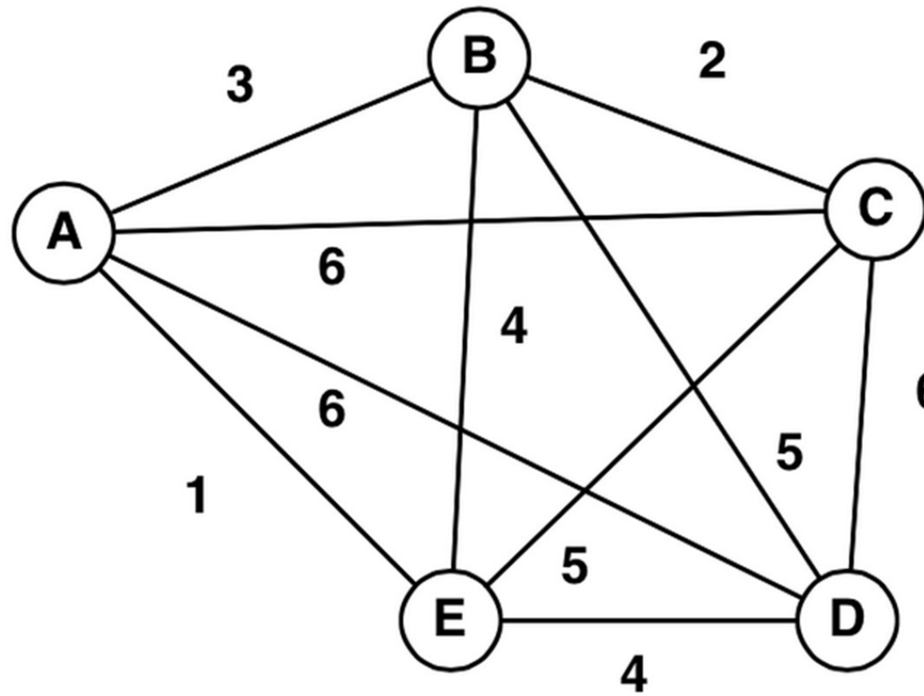
Graph

A graph is determined as a mathematical structure that represents a particular function by connecting a set of points. It is used to create a pair wise relationship between objects. The graph is made up of vertices (nodes) that are connected by the edges (lines).

The applications of the linear graph are used not only in Maths but also in other fields such as Computer Science, Physics and Chemistry, Linguistics, Biology, etc. In real-life also the best example of graph structure is GPS, where you can track the path or know the direction of the road.

What is Graph



In Mathematics, a graph is a pictorial representation of any data in an organized manner. The graph shows the relationship between variable quantities between the vertices (nodes) and edges (lines).

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A graph is denoted as a pair G(V, E). Where V represents the finite set of vertices and E represents the finite set of edges.

Therefore, we can say a graph includes non-empty set of vertices V and set of edges E.

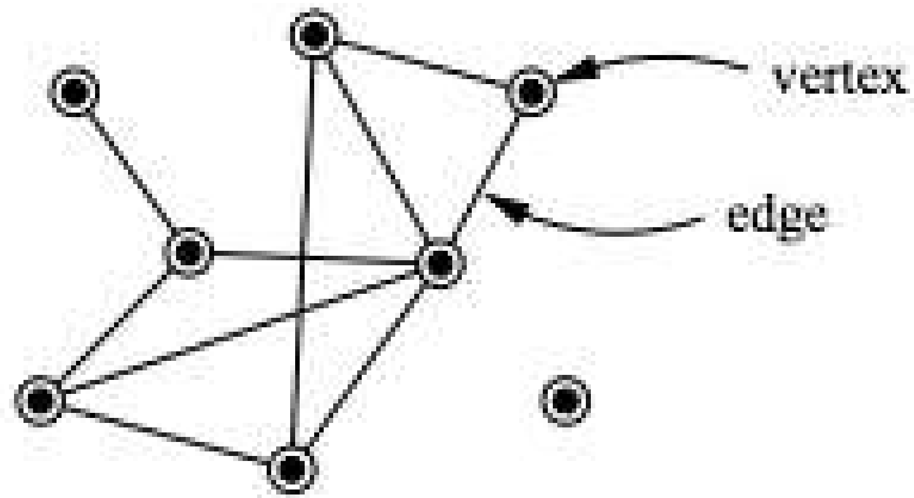
Example

Suppose, a Graph G=(V,E), where

Vertices, V={a,b,c,d}

Edges, E={{a,b},{a,c},{b,c},{c,d}}

Vertices: Vertices are a point where one or more straight line or curve meet.

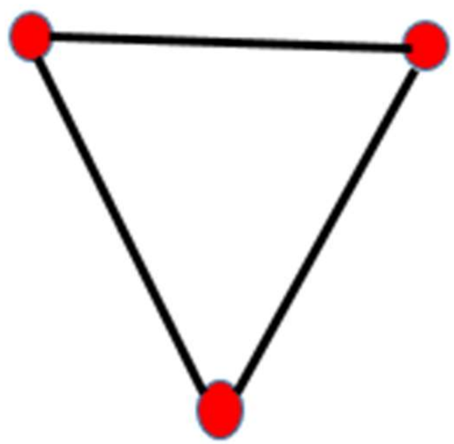


Or, A vertices V whose elements are called vertices points or node of the Graph G.

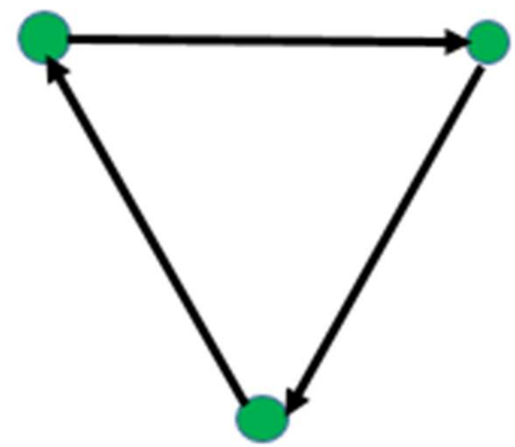
Edges: Edges are a line or curve segment that join two or more vertices. Or, A set of un-directed pairs of distinct vertices called Edges of the Graph G

Types of Graph

The graphs are basically of two types, directed and undirected.



Undirected Graph



Directed Graph

Directed Graph

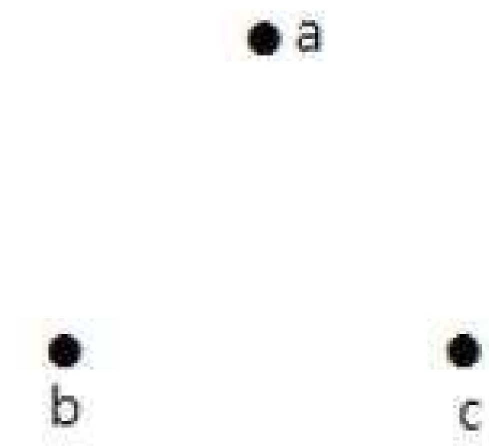
In graph theory, a directed graph is a graph made up of a set of vertices connected by edges, in which the edges have a direction associated with them.

Undirected Graph

The undirected graph is defined as a graph where the set of nodes are connected together, in which all the edges are bidirectional. Sometimes, this type of graph is known as the undirected network.

Other types of graphs

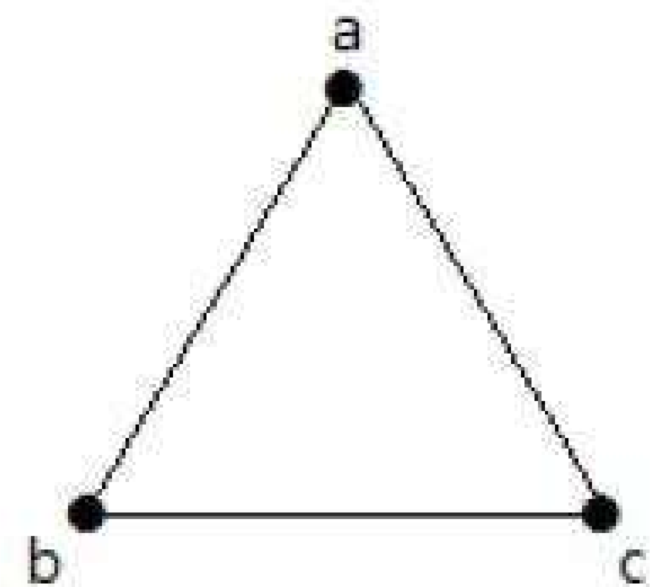
Null Graph: A graph that does not have edges.



Trivial Graph

A graph with only one vertex is called a Trivial Graph.

Simple graph: A graph that is undirected and does not have any loops or multiple edges. A graph with no loops and no parallel edges is called a simple graph.



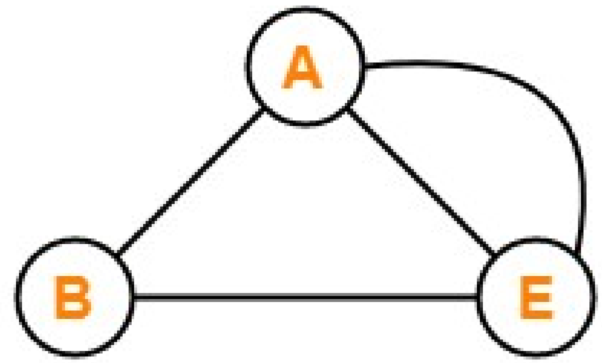
The maximum number of edges with n=3 vertices

$$
\begin{aligned}
{}^nC_2 &= n(n-1)/2 \\
&= 3(3-1)/2 \\
&= 6/2 \\
&= 3 \text{ edges}
\end{aligned}
$$

The maximum number of simple graphs with n=3 vertices

$$
\begin{aligned}
2^{{}^nC_2} &= 2^{n(n-1)/2} \\
&= 2^{3(3-1)/2} \\
&= 2^3 \\
&= 8
\end{aligned}
$$

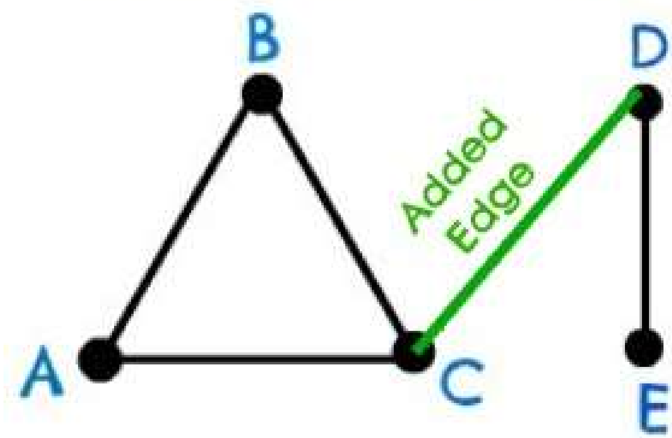
Multi graph: A graph with multiple edges between the same set of vertices. It has loops formed.



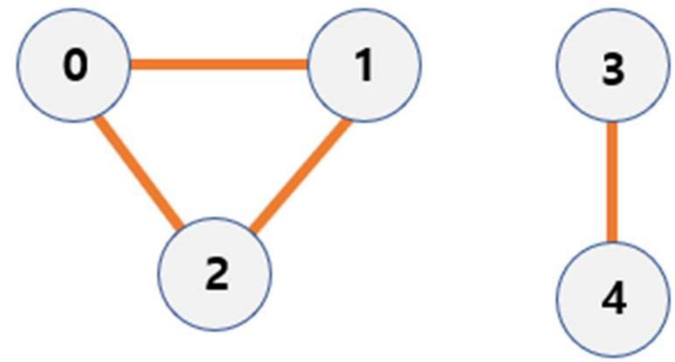
Example of Multi Graph

Connected graph: A graph where any two vertices are connected by a path. (Vertex not repeated, Edge not repeated.)

Connected Graph

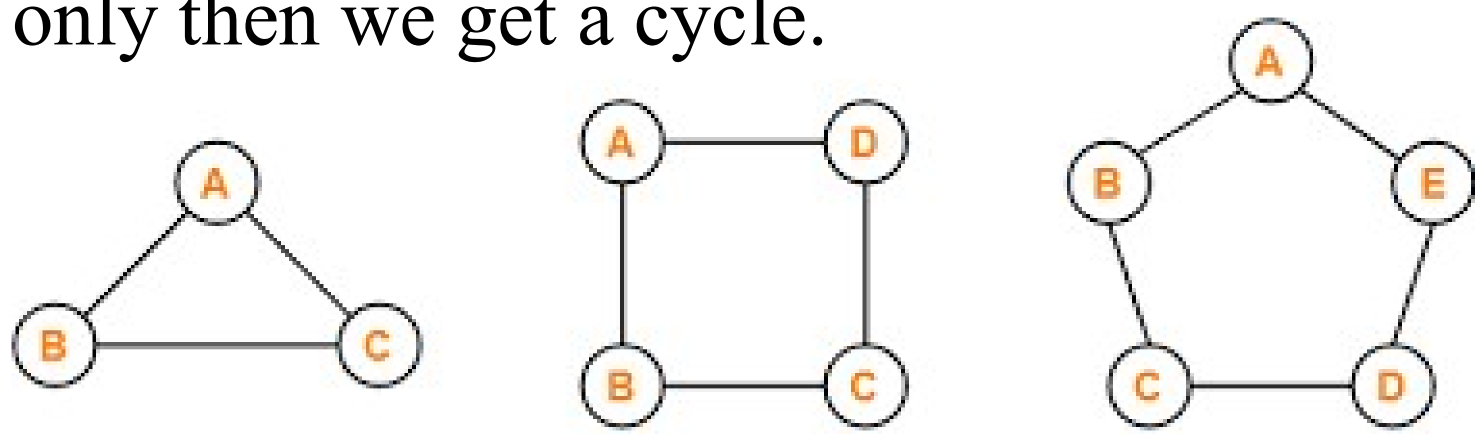


Disconnected graph: A graph where any two vertices or nodes are disconnected by a path.



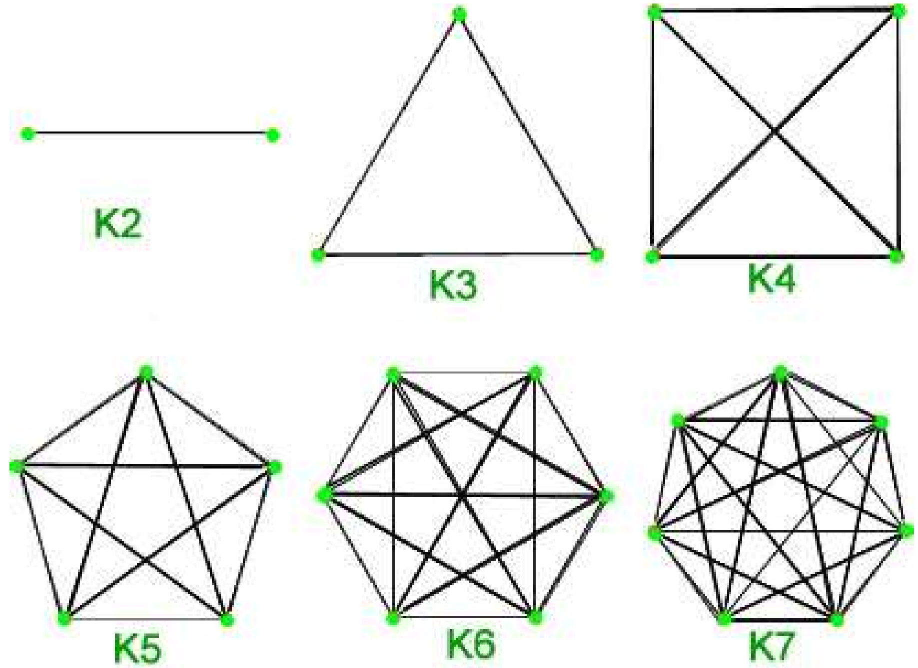
Disconnected Graph

Cycle Graph: A graph that completes a cycle. Traversing a graph such that we do not repeat a vertex nor we repeat a edge but the starting and ending vertex must be same i.e. we can repeat starting and ending vertex only then we get a cycle.

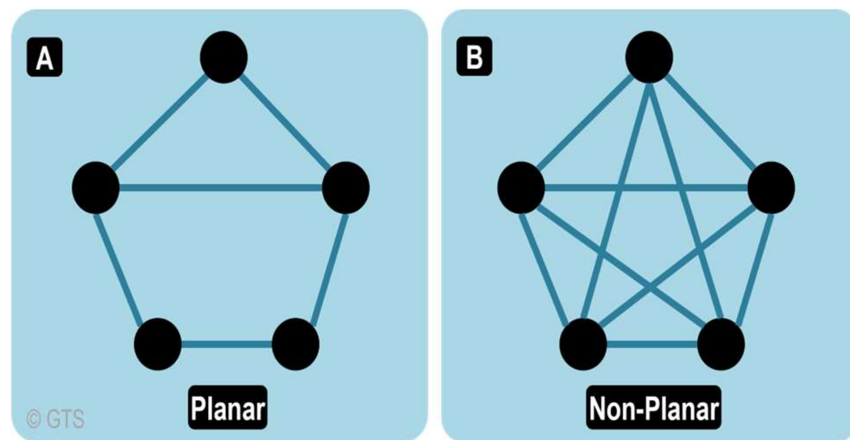


Examples of Cycle Graph

Complete Graph: When each pair of vertices are connected by an edge then such graph is called a complete graph.

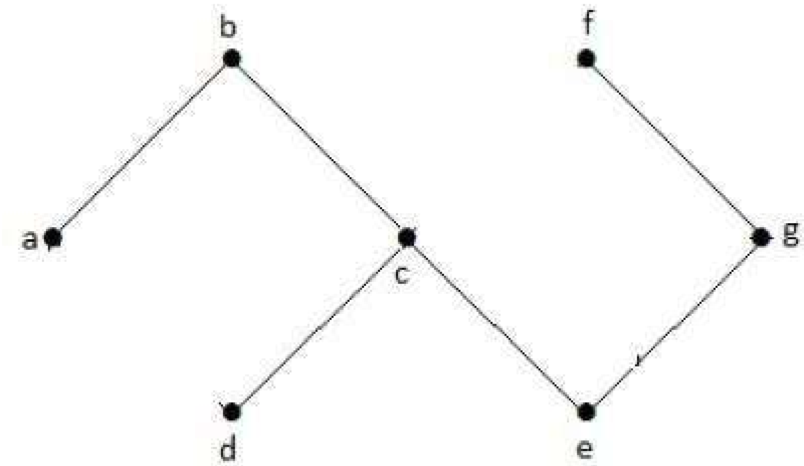


Planar graph: When no two edges of a graph intersect and are all the vertices and edges are drawn in a single plane, then such a graph is called a planar graph



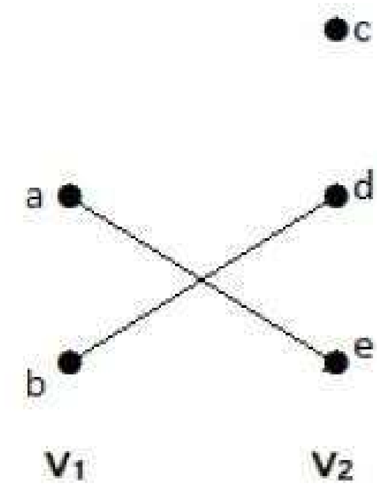
Acyclic Graph

A graph with no cycles is called an acyclic graph.



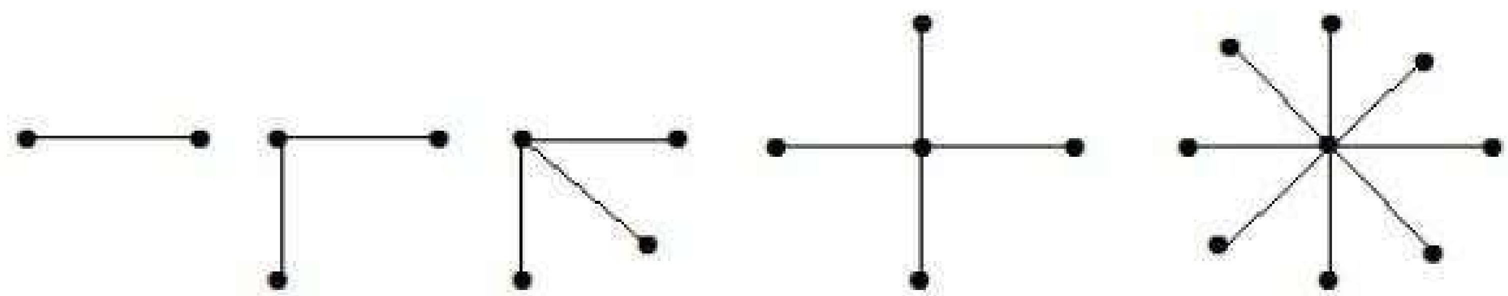
Bipartite Graph

A simple graph G = (V, E) with vertex partition V = {V1, V2} is called a bipartite graph if every edge of E joins a vertex in V1 to a vertex in V2.



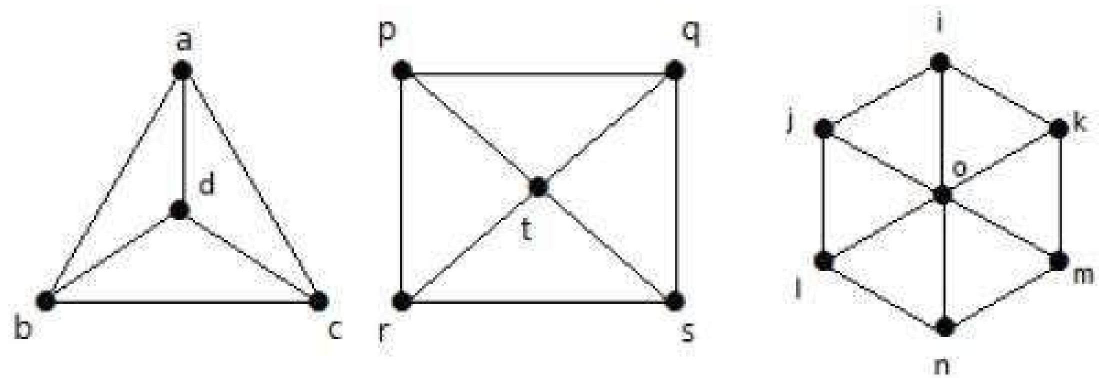
Star Graph

A star graph is a complete bipartite graph if a single vertex belongs to one set and all the remaining vertices belong to the other set.



Wheel graph

A wheel graph is a graph formed by connecting a single universal vertex to all vertices of a cycle.



I(W_4) II(W_5) III(W_7)

Properties of Graph

The starting point of the network is known as **root.** When the same types of nodes are connected to one another, then the graph is known as an **assortative** graph, else it is called a **disassortative** graph.

A cycle graph is said to be a graph that has a single cycle. When all the pairs of nodes are connected by a single edge it forms a complete graph.

A graph is said to be in symmetry when each pair of vertices or nodes are connected in the same direction or in the reverse direction. When a graph has a single graph, it is a path graph.

Isomorphic graphs

A graph can exist in different forms having the same number of vertices, edges, and also the same edge connectivity. Such graphs are called isomorphic graphs.

Two graph which contain the same number of graph vertices connected in the same way.

Conditions for graph isomorphism

Any two graphs will be known as isomorphism if they satisfy the following four conditions:

- There will be an equal number of vertices in the given graphs.

- There will be an equal number of edges in the given graphs.

- There will be an equal amount of degree sequence in the given graphs.

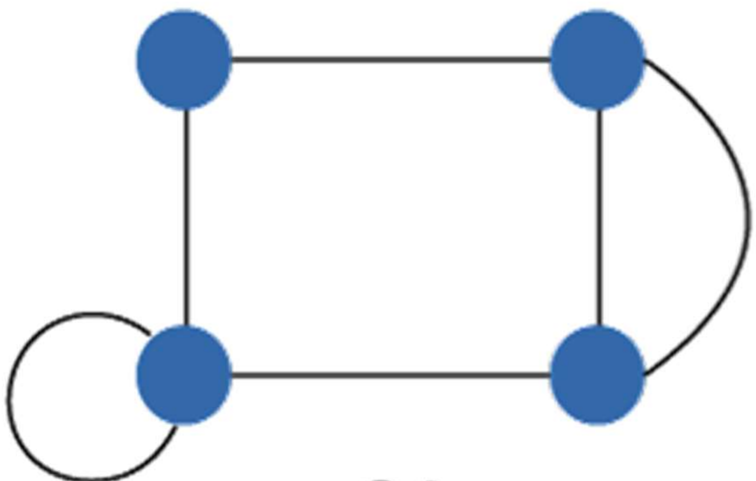
- If the first graph is forming a cycle of length k with the help of vertices {v1, v2, v3, …. vk}, then another graph must also form the same cycle of the same length k with the help of vertices {v1, v2, v3, …. vk}.

Sufficient Conditions for Graph isomorphism

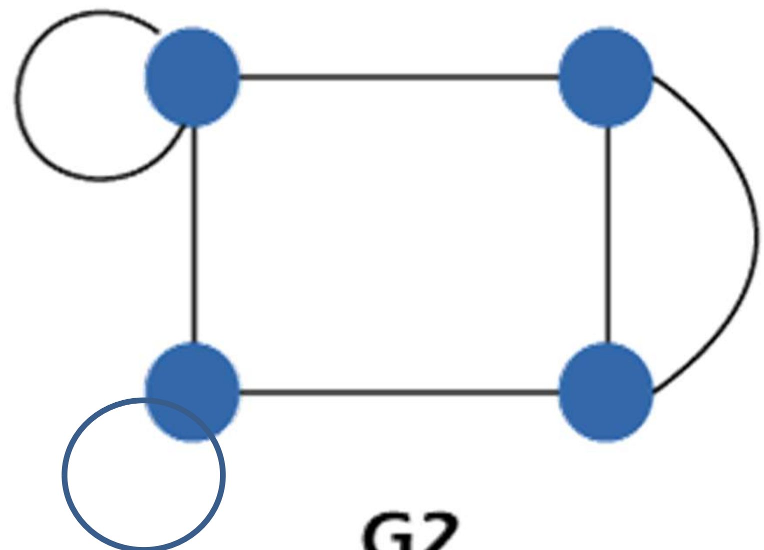
If we want to prove that any two graphs are isomorphism, there are some sufficient conditions which we will provide us guarantee that the two graphs are surely isomorphism. When the two graphs are successfully cleared all the above four conditions, only then we will check those graphs to sufficient conditions

- If the complement graphs of both the graphs are isomorphism, then these graphs will surely be an isomorphism.
- If the adjacent matrices of both the graphs are the same, then these graphs will be an isomorphism.
- If the corresponding graphs of two graphs are obtained with the help of deleting some vertices of one graph, and their corresponding images in other images are isomorphism, only then these graphs will not be an isomorphism.

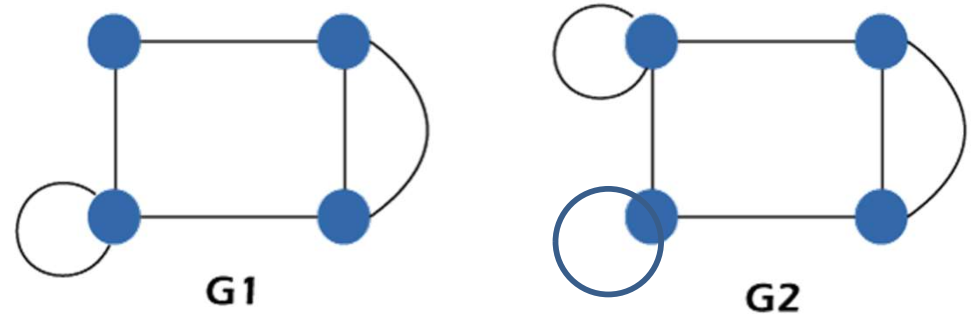
Shown that whether the following graphs are isomorphic or not.



G1



G2



Condition 1:

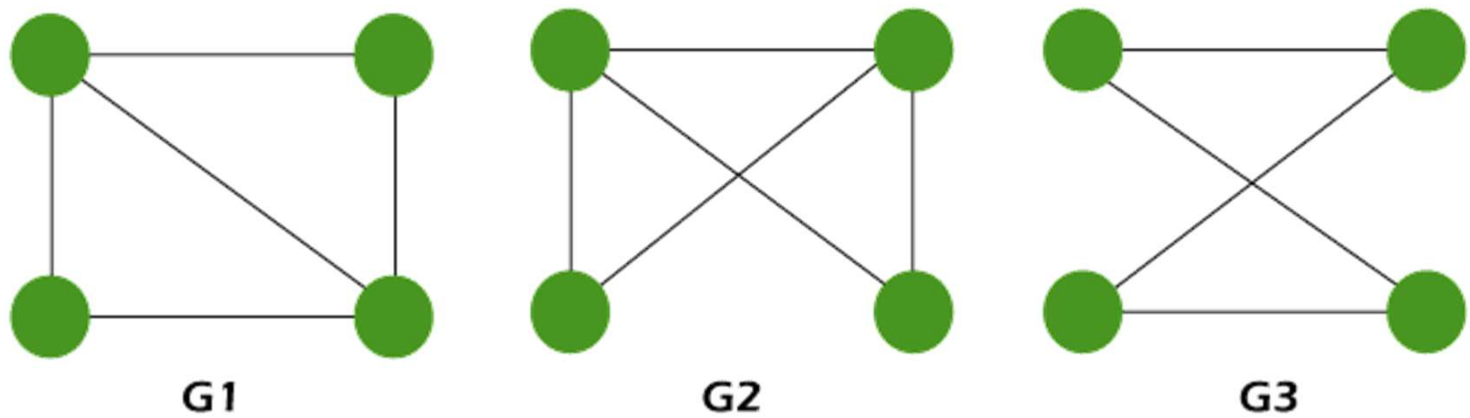
In graph 1, there is a total 4 number of vertices, i.e., G1 = 4.
In graph 2, there is a total 4 number of vertices, i.e., G2 = 4.
Here,
There are an equal number of vertices in both graphs G1 and G2. So these graphs satisfy **condition 1**

Condition 2:

In graph 1, there is a total 5 number of edges, i.e., G1 = 5.
In graph 2, there is a total 6 number of edges, i.e., G2 = 6.
Here,

There does not have an equal number of edges in both graphs G1 and G2. So these graphs do not satisfy **condition 2**. Since, these graphs violate **condition 2.** So these graphs are not an isomorphism.

∴ Graph G1 and graph G2 are not isomorphism graphs.



Condition 1:

In graph 1, there is a total 4 number of vertices, i.e., G1 = 4.
In graph 2, there is a total 4 number of vertices, i.e., G2 = 4.
In graph 3, there is a total 4 number of vertices, i.e., G3 = 4.

Condition 2:

In graph 1, there is a total 5 number of edges, i.e., G1 = 5.
In graph 2, there is a total 5 number of edges, i.e., G2 = 5.
In graph 3, there is a total 4 number of edges, i.e., G2 = 4.

Since, the graphs (G1, G2) and G3 violate condition 2. So we can say that these graphs are not an isomorphism.

∴ Graphs G1 and G2 may be an isomorphism.

Condition 3:

In the graph 1, the degree of sequence s is {2, 2, 3, 3}, i.e.,

G1 = {2, 2, 3, 3}.

In the graph 2, the degree of sequence s is {2, 2, 3, 3}, i.e.,

G2 = {2, 2, 3, 3}.

Here

There are an equal number of degree sequences in both graphs G1 and G2. So these graphs satisfy condition 3. Now we will check the fourth condition.

Condition 4:

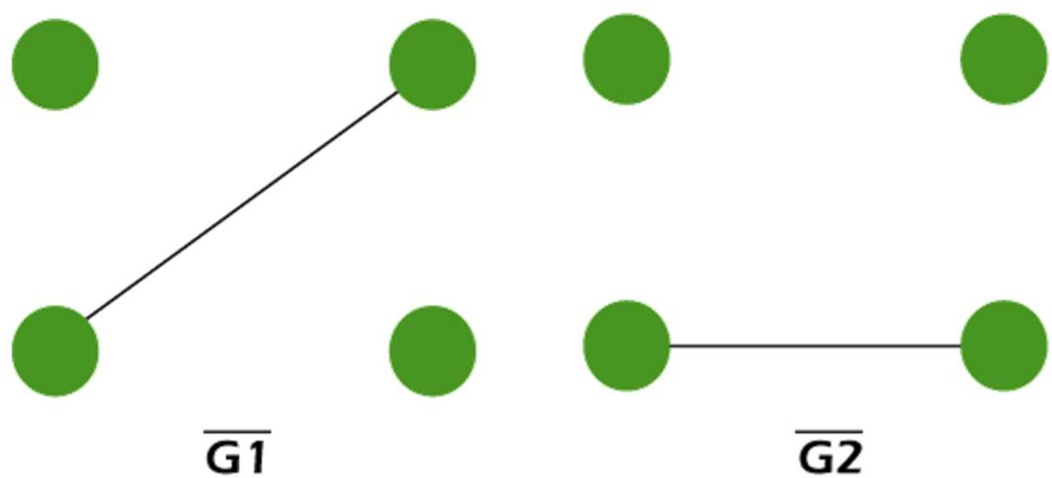
Graph G1 forms a cycle of length 3 with the help of vertices {2, 3, 3}.
Graph G2 also forms a cycle of length 3 with the help of vertices {2, 3, 3}.

Here,
It shows that both the graphs contain the same cycle because both graphs G1 and G2 are forming a cycle of length 3 with the help of vertices {2, 3, 3}. So these graphs satisfy condition 4.

Thus,

The graphs G1 and G2 satisfy all the above four necessary conditions. So G1 and G2 may be an isomorphism.

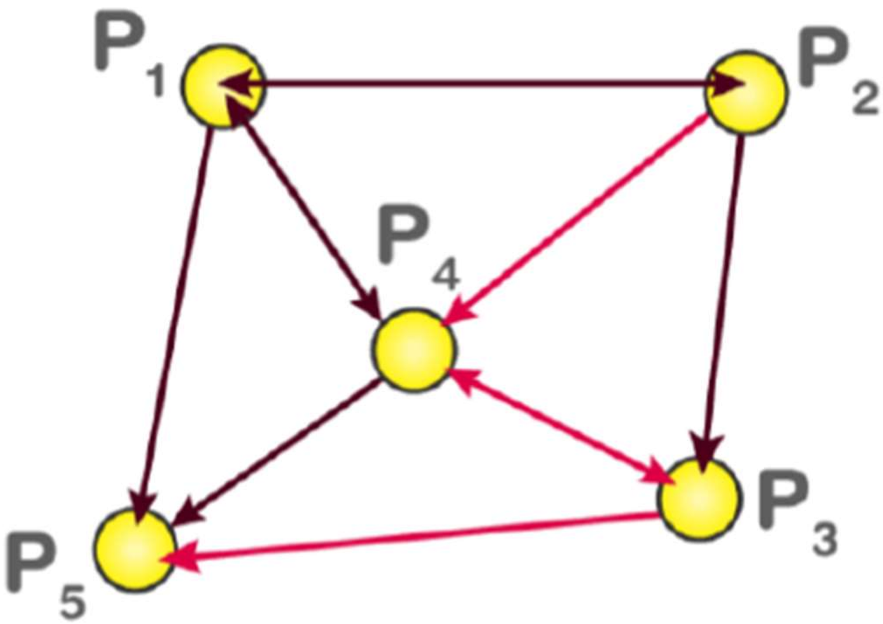
As we have learned if the complement graphs of both graphs are isomorphism, the two graphs will surely be an isomorphism. So we will draw the complement graphs of G1 and G2, which are described as follows:



In the above complement graphs of G1 and G2, we can see that both the graphs are isomorphism.

The **adjacency matrix**, also called the **connection matrix**, is a matrix containing rows and columns which is used to represent a simple labelled graph, with 0 or 1 in the position of (V_i , V_j) according to the condition whether V_i and V_j are adjacent or not. It is a compact way to represent the finite graph containing n vertices of a m x m matrix M. Sometimes adjacency matrix is also called as **vertex matrix** and it is defined in the general form as

$$\begin{Bmatrix} 1 & if\, P_i \rightarrow P_j \\ 0 & otherwise \end{Bmatrix}$$



i

j	0	1	2	3
0	0	1	1	1
1	1	0	1	0
2	1	1	0	0
3	1	0	0	0

$$\text{The adjacency matrix } = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

END